

Franklin Heng

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CV/ML engineer building solutions across **healthcare**, **climate**, and **robotics**. Deployed and built solutions from 0 → 1.

WORK EXPERIENCE

Applied ML Engineer, Make The Dot

September 2025 – January 2026

- **Built and optimized an end-to-end stable diffusion pipeline** for generating photo-realistic renders of clothing from designer sketches. Pipeline leverages open-source models (Juggernaut X v10), image preprocessing for control inputs, and ControlNet++/ControlNet Union model. [\[view product overview\]](#).
- **Fine-tuned Juggernaut X** using **LoRA** for accurate fabric/materials rendering. Merged LoRA weights to blend multiple fabric types and generate new fabrics during inference.
- **Managed full ML lifecycle** from data collection, training, model serving/inference, and monitoring at scale.
- **Compressed ML pipeline with TensorRT** and served on **NVIDIA Triton Inference Server** for scalable and fast inference.

Computer Vision Engineer, Running Tide

March 2022 – June 2024

- **Built a computer vision pipeline** to detect biomaterials from offshore images captured by our **custom-built camera buoys** and track sinking rates over time. Mask-RCNN was used for segmentation (Tensorflow), and the sinking rate was computed by measuring the percentage of biomaterials floating (OpenCV, Scikit-Image). **I was responsible for data collection, data labeling, model training/evaluation, and deployment.** I containerized the pipeline using Docker and deployed it to production using Google Cloud Composer and Airflow. [\[view abstract\]](#)
- **Fine-tuned Faster-RCNN** model to detect and count shellfish on our **robotic shellfish farm** with the requirement of **90%+ accuracy** and processing **speed of .125s per image**, and developed post-processing scripts to fine-tune detection bounding boxes (Tensorflow, OpenCV). Each image could contain ~1000 shellfish and as small as 2x2 pixels.
- **Fine-tuned SAM Vision Transformer (ViT)** model to identify and segment macroalgae from offshore images with **90% (IoU) accuracy**, and trained a **linear regression model** to predict weight within **1 gram of error**.
- **Generated high-resolution 3D reconstructions and point clouds** (COLMAP, Metashape) of macroalgae from multiview images and **computed 3D measurements** (Open3D). [\[view product overview\]](#) [\[view abstract\]](#)

Computer Vision Engineer & Co-founder, Breathily (funded by UCSF)

March 2020 – March 2022

- Co-founded a startup to develop a solution that enables patients with physically debilitating diseases, such as ALS, to measure their lung function using RGB-D cameras. [\[view video demo\]](#)
- **Developed a computer vision pipeline** to compute lung function values from data captured by an RGB-D camera (Intel RealSense). Pipeline involved **pose estimation** (MediaPipe), **RGB-D image processing** (OpenCV), 1-D signal analysis (SciPy), 3D mesh estimation (Pixel2Mesh), and CNN+LSTM networks (Keras) [\[view product overview\]](#)
- **Co-inventor of U.S. patent** application entitled “Methods for Pulmonary Function Testing With Machine Learning Analysis and Systems for Same” filed February 2, 2022. [\[view patent\]](#)
- **Co-led an IRB-approved research study** that enabled our team to capture data and test our device on live patient visits at UCSF Pulmonary Function Lab.
- **Secured funding** from UCSF Catalyst Program, NSF National I-Corps Program, and UC Launch Accelerator Program.

Computer Vision Scientist, UC San Francisco Hospital (UCSF)

November 2018 – March 2020

Fahy Lab (June 2019 - March 2020)

- **Fine-tuned Mask-RCNN model to segment and locate mucus plugs in lung airways** from chest CT scans.
- Leveraged **fuzzy clustering (GK-clustering)** to cluster mucus from the lung parenchyma on a volumetric basis. Using the 3D clusters, I computed length and width measurements using **principal component analysis** and **cylinder fitting**. Triangulation and a **3D mesh** of the mucus surface were generated using **marching cubes** for further visualization (Scikit-Image, MATLAB).
- Segmented central airways from chest CT scans using **region growing** and **CNN methods**. This enabled our team to map the mucus plug to specific branches within the lung airway tree. With the combination of mucus plug location, airway size, we **developed a novel method to measure airway resistance** and obstruction.
- **Published paper in JCI Insight** [\[view publication\]](#) and multiple abstracts.

Abbasi Lab (November 2019 - June 2020)

- Leveraged **non-negative matrix factorization (NMF)**, correlation, and **linear regression** to predict levels of engagement from EEG signals.
- Pre-processed EEG signals by de-noising, applying Morelet wavelet transformation, and analyzed spectrograms by generating topography maps to visualize evoked locations of EEG.
- [\[View PowerPoint\]](#) outlining the project and methodology

Sohn Lab (November 2018 - June 2019)

- **Designed a custom 3D CNN architecture** to predict perceived brain age from brain CT scans. The difference between predicted perceived age and actual (true) age was used to predict a panel of systematic diseases and conditions.
- **Developed a preprocessing method** that isolated soft tissue (via inversion), resampling to 1mm thickness, normalization, and resizing. Furthermore, region segmentation was applied, which involves dividing brain slices into 8 equally sized regions before feeding them into the model.
- **Generalized linear model and logit function** were created for **univariate and multivariate analysis** of correlation between predicted and true age difference and systemic diseases/conditions (hypertension, diabetes, polysubstance abuse, etc.)

Computer Vision Researcher, UC Berkeley

May 2017 – January 2019

- **Developed a computer vision and image processing pipeline** to measure the width of plant stems from videos captured by a moving **agricultural robotic system**.
- **Faster-RCNN model** was used for **plant stem detection**. Edge detection, morphological operation, and RANSAC were performed sequentially for final width estimation.
- [\[View publication\]](#) for further details of research and methodology.

Full-Stack AI Engineering Projects

[FOIA Fluent](#), Data Science for Social Good (NYCxDSGG)

March 2026 - Present

- Built the entire website end-to-end (FastAPI, Next.js 14, Supabase), which drafts, tracks, and analyzes FOIA / FOIL / CPRA / PIA requests [\[view repo here\]](#).
- Developed an AI chat assistant with 7 custom tools and a 4-tier response pipeline (local lookup → trusted-domain search → deep research agent → fallback)
- Built Data Hub/intelligence layer, aggregating and visualizing FOIA outcomes across 1,600+ federal agencies and 54 state jurisdictions with daily refresh from 20 verified sources.

SKILLS

Computer Vision: Tensorflow, PyTorch, Keras, Hugging Face, OpenCV, SciKit-Image, Google Vertex AI

3D Vision: COLMAP, Metashape, Open3D, RGB-D sensors, Lidar

Software: Docker, Google Cloud Composer, Apache Airflow, TensorRT, Agentic tooling & IDEs

Languages: Python, MATLAB, C++

EDUCATION

B.A, Computer Science, University of California, Berkeley

SELECTED PUBLICATIONS, PATENTS, & ABSTRACTS

- [1] B. Huang, ..., **F. Heng**, ..., J. Fahy, "Mucus plugs in proximal airway generations are consequential for airflow limitation in asthma" JCI Insight, December 2023 **[Computer Vision, Health]**
- [2] **F. Heng**, J. Belanger, M. Horowitz, K. Brown, C. Payne, R. Masalia "Computer-vision based phenotyping approaches in the brown macroalgae, Saccharina Latissima" NAPPN National Conference 2022, St. Louis, Missouri, October 2022 **[Computer Vision, Climate]**
- [3] **F. Heng**, X. Orain, "Methods for Pulmonary Function Testing With Machine Learning and Systems for Same", US Patent, filed February 2022, Patent Pending **[Computer Vision, Health]**
- [4] A. Sahiner, **F. Heng**, A. Balamurugan, and A. Zakhori, "In Situ Width Estimation of Biofuel Plant Stems" Electronic Imaging Conference 2019, Burlingame, California, January 2019 **[Computer Vision, Robotics]**